Start with one trustworthy result

What we are building

A researcher opens Analysis, chooses a protein run, and gets a saved RMSD curve with units, settings and warnings. Later, Run Simulation will start the lab's prepared water/peroxide exercise and display live progress. These are separate workflows: analysis reads existing coordinates; simulation generates new ones.

The pieces, in plain language

The TUI is the terminal screen built with Textual. The backend is the Python code that performs an analysis without needing that screen. GROMACS is the scientific program the backend calls. A runner is our small Python component that starts GROMACS and reports what happened. It does not calculate molecular properties.

One example from click to file

Click Run RMSD. The form makes a request (files, groups and time window). A background worker performs it while the screen remains responsive. The backend asks the runner to start gmx rms. GROMACS writes a numerical table; the backend reads it and returns an AnalysisResult: data, statistics, file locations and information about how they were produced.

What belongs where

Keep the existing environment detection and monitor. Bring the supplied analysis_backend source and tests into the project, recording their origin. Leave its virtual environment (a machine-specific Python installation), cached files and large lab trajectories outside the repository. Keep the PripA research pipeline separate: it answers additional pore-transport questions.

Milestone order

Verify one calculation without the screen. Build only enough shared execution code to run it reliably. Connect RMSD end to end, then add RMSF and radius of gyration. Reuse the runner and monitor for a short prepared simulation.

1 / Establish a scientific baseline

Choose a manageable example

Use a short, known-good protein trajectory segment with matching topology. A trajectory (.xtc or .trr) stores coordinates at successive times; a frame is one snapshot. A .tpr describes the run and its atoms. An optional .ndx lists groups of atoms. Matching means the atoms correspond in number and order; a matching filename alone proves little.

Use the correct data

The student exercise is a solvent box: useful for monitoring, density and launching, but it has no protein backbone. PripA has protein, but its full-system coarse data and solvent/protein fine subset have different atom lists. The fine data only cover the last roughly 32.5 ns of a 500 ns run. Start with one validated segment; show its actual coverage.

Make a reference result

In one clean environment run the supplied backend tests, then run gmx rms directly on the chosen segment. Save the executable version, inputs, reference structure, fit settings, groups, time window and output. Repeat through run_rmsd using identical choices. Check the time column, number of rows and values within the precision of the saved table.

Explain the options

RMSD measures distance from a reference structure. The fit group aligns snapshots to remove overall movement; the calculation group is the set measured after alignment. Record how jumps across periodic box boundaries were handled. The supplied RMSD backend reports time in nanoseconds and distances in nanometres. The lab reference may use angstroms: 1 nm = 10 angstroms.

Done when

Document one reproducible command and reference table, plus one expected failure such as a missing group. Verify the required Python libraries, including NumPy and pandas; keep optional plotting dependencies separate. The dashboard's earlier 28 passing tests concern the existing dashboard, not this new backend.

2 / Connect RMSD to the screen

Runner: one way to start a command

Use the executable already verified by Setup. For the first demonstration, launch both Python and GROMACS inside WSL (Windows Subsystem for Linux). For Windows-native launch, translate file paths into the selected WSL distribution and test paths containing spaces. Use the same runner for version checks and calculations, including RMSF's separate execution helper.

Job: one user request with a history

Give each request its own output folder and state: queued, running, succeeded, failed or cancelled. Capture GROMACS messages in a log, pass the chosen groups as input and keep the process handle for cancellation. Replace the backend's fixed five-minute calculation timeout with a suitable job policy. Update the screen from the worker; show elapsed time when true percentage is unknown.

Form and validation

Ask for topology, trajectory, optional index, fit and measurement groups, time window and destination. Show detected groups and units. Check that inputs are readable, the group exists for this system and outputs can be written. Explain incompatible or unverified atom mappings. Prevent double-clicks from launching duplicate work.

Result screen

Show the curve, actual time range and numerical summary. Save XVG (GROMACS numerical output with plot labels), CSV (a spreadsheet-readable table), execution log and a metadata file describing the inputs, selections and version. Distinguish a calculation failure from a statistical warning. Repeated runs must never erase a previous result.

Done when

The same example produces the baseline curve from the TUI. A missing file produces an actionable message; cancellation stops the calculation; a second run has a different results folder. Verify results reopen without needing the original screen.

3 / Add two structural measurements

RMSF: which parts move most?

Root mean square fluctuation describes motion around an average position. A per-residue result plots fluctuation against residue identity, not time. Give this method its own group and per-residue choices; retain chain identity where available. Optional window comparisons split the time range and rerun calculations: explain the extra work before starting.

Radius of gyration: how compact is it?

Rg describes how far the selected atoms are distributed from their centre, with weighting defined by the method. Plot total Rg against time. The output may also contain directional components; use labels and the supplied parser to identify the total. Check the GROMACS version and actual output units before displaying it.

Reuse and verify

Reuse the file picker, job worker, runner, warnings and result folder. Keep each method's scientific settings and chart labels distinct. Compare each result with an equivalent direct GROMACS run. For comparison with the lab's QC figures, match reference, atom selections, weighting, alignment, preprocessing, time interval and units first.

Keep the first release small

One run at a time is enough initially. Use existing CSV/XVG outputs and a simple plot or HTML export. Introduce multi-run comparisons, scheduling and specialised pore analyses only after researchers can reproduce the three basic measurements.

4 / Launch a prepared simulation

First screen

Offer the student water/peroxide template, a run name, destination, Short test or Full exercise, and Auto/CPU/GPU. CPU is the general processor; a compatible GPU can accelerate selected calculations. Show the chosen executable and mode. Keep the lab's physical settings as a preset. Building an arbitrary system from a protein structure is a separate future project.

What the files mean

The .gro contains starting coordinates; .top and included .itp files describe molecules and forces; .mdp contains settings such as timestep and number of steps. Copy the full template, including the force-field folder, into a fresh run directory. Change only the copied settings for a short test and show exactly which stages are shortened.

Stages the runner performs

For each stage, grompp checks and combines inputs into a .tpr; mdrun executes it. Energy minimisation relaxes bad contacts. Equilibration lets the box adjust under target conditions. Production generates the trajectory for measurements. Use each stage's required outputs as the next stage's inputs, including the equilibration checkpoint where required.

Monitor, stop and resume

Show the active stage and feed its log into the existing monitor for step, time, speed and estimated finish. Save standard output and errors. Stop on a failed stage and show the relevant warning; investigate the exercise's -maxwarn 2 before automating it. A checkpoint (.cpt) saves simulation state. Resume only the matching existing run; Start new creates another directory.

Acceptance demo

On CPU, complete a short copied exercise and confirm all three stages, logs and final files. Interrupt production, restart from its checkpoint and verify time advances. Test GPU separately on compatible lab hardware. Confirm the template is unchanged and an invalid input stops before production. A short run demonstrates the software; use the full protocol for scientific results.